Introduction to Economics

Class 7 — Elasticity

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Plan for today

Warm-up; Class 6, one more look
Olive oil abroad: a third less, the same money
Elasticity: the number, the midpoint
One line, many elasticities; in pairs
What makes demand elastic
Price × quantity: the box
Break
Topic fixed; Problem set 1 out
The olive oil answered; raise or cut?
Income and cross-price elasticities
Supply: when sellers can respond
In one line; before Monday 19 October

Slide 1 Title

Preview version. The full slides and handout are posted after class.

Class 6 settled which way a price moves when a curve shifts; Class 7 asks how much. One number does the work: elasticity, the percentage change in quantity over the percentage change in price. The first hour defines it, computes it between two points and along a line, and uses it to say what a price change does to a seller's money. The second hour pays off the opening puzzle, adds income and cross-price elasticities, and gives sellers an elasticity of their own. The thread is a real pair of numbers: Spain sent a third less olive oil abroad in 2023 and got the same money back.

Slide 2 Plan for today

The class at a glance. A short warm-up on two moves from Classes 5 and 6, then a last look at Class 6's teaser, then a real pair of numbers that only today's idea explains. The definition of elasticity, the method for computing it between two points, and a line that is elastic at one end and inelastic at the other; then what a price change does to the money in a market. After the break the puzzle gets its answer, two more elasticities whose sign is the point, and the sellers' side, where time is the key.

Two halves of fifty minutes with a ten-minute break between them.

Warm-up: which curve moves, and which way?

Coach tickets, Barcelona–Valencia: for students, an inferior good

1. Students' incomes fall
2. Car-sharing rides get dearer
3. The coach fare falls
4. Diesel gets dearer

In pairs, two minutes; then hands per item.

Slide 3 Warm-up: which curve moves, and which way?

A warm-up on telling a shift of a curve from a move along it. Before class, decide for each of the four events which curve moves, demand or supply, and which way, or whether buyers only move along a curve. The slide says that for students the coach is an inferior good: use it.

Revealed in class.

Warm-up: what does the whole market buy?

Padel-court hours a month

price	Ana	Biel	the market
10 euros	5	3	
20 euros	3	1	

Carla joins: 2 hours at 10 euros, 1 at 20. The market curve?

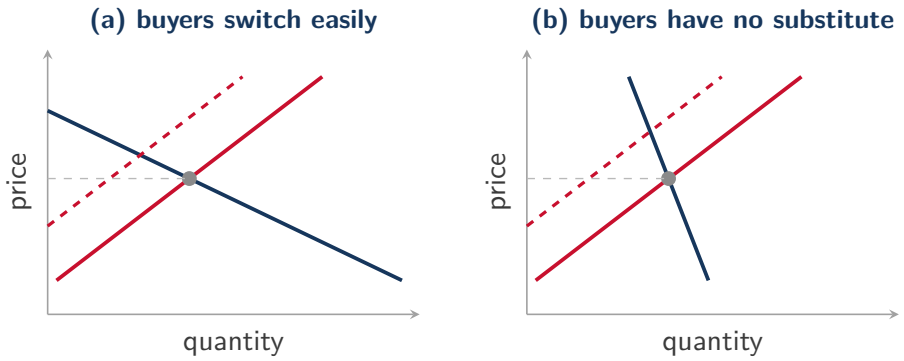
In pairs, one minute; then hands.

Slide 4 Warm-up: what does the whole market buy?

A warm-up on the market's demand. Before class, find what the two buyers take together at each price, and say what happens to the market's curve when a third buyer joins with 2 hours at 10 euros and 1 at 20.

Revealed in class.

Class 6 again: the same shift — where does the price jump more?



Slide 5 Class 6 again: the same shift, where does the price jump more?

Class 6's last picture, as a question. The same supply shift in two markets: in (a) buyers switch easily and the demand line is flat; in (b) they have no substitute and it is steep. The new crossings are not drawn: find them where the dashed supply meets each demand line. Before class, decide in which panel the price jumps more and in which the quantity falls more, and say in one sentence what differs between the two markets, given that the shift is the same.

Revealed in class.

Olive oil abroad in 2023: a third less oil, the same money

1.07 million tonnes → **0.68 million tonnes**

shipped abroad, 2022 → 2023

4.10 bn € → **4.15 bn €**

what came back, 2022 → 2023

What does that say about the buyers abroad?

Spain's olive-oil exports, customs data (AEAT), in MAPA's 2023 product report; 2023 provisional.

Slide 6 Olive oil abroad in 2023: a third less oil, the same money

The hook, from the customs office. Spain shipped 1.07 million tonnes of olive oil abroad in 2022 and 0.68 million in 2023, after the dry year of Class 6: about a third less. The money that came back was 4.10 billion euros, then 4.15 billion: the same. A third less of something and the same money is odd only until you ask how the buyers abroad reacted to the price. The answer waits for the second half; the slide is here so that the question is in your head while the tool is built.

Elasticity: how much the quantity moves, in percentages

$$\text{price elasticity of demand} = \frac{\%\text{ change in quantity demanded}}{\%\text{ change in price}}$$

- > 1 **elastic**: the quantity moves more than the price
- < 1 **inelastic**: the quantity moves less than the price
- $= 1$ **unit elastic**: the same

Percentages, so cups and tonnes compare. The minus sign is dropped.

Slide 7 Elasticity: how much the quantity moves, in percentages

The definition in words: the price elasticity of demand is the percentage change in the quantity demanded divided by the percentage change in the price. Percentages, because forty fewer cups and 0.4 million tonnes less cannot be compared, while two percentages can. Along a demand curve the quantity falls when the price rises, so the ratio is negative; the sign is dropped and only its size is read.

Elastic means stretchy: a small pull on the price and the quantity moves a lot, more than one per cent for each per cent of price. Inelastic is the reverse; unit elastic is the border, where the two percentages are equal.

Two points, one number: percentages of the midpoint

The neighbourhood's cups: 2.00 → 2.50, 160 → 120 a day

computed from	price	cups	elasticity
the start, 2.00	+25 %	−25 %	1.0
the start, 2.50	−20 %	+33 %	1.67
the midpoint	**+22 %**	**−29 %**	**1.29**

$$\frac{(Q_2 - Q_1) \,/\, \frac{1}{2}(Q_1 + Q_2)}{(P_2 - P_1) \,/\, \frac{1}{2}(P_1 + P_2)}$$

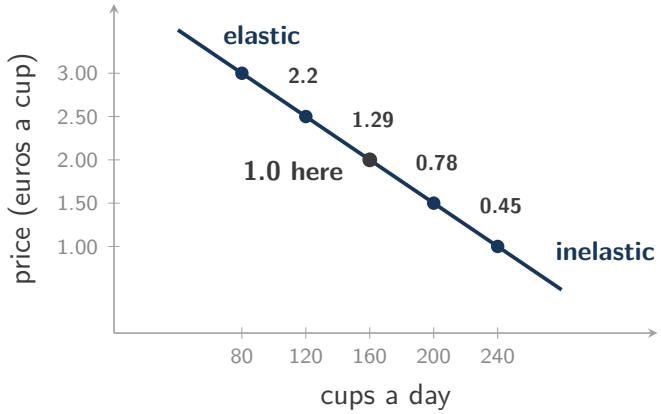
Divide each change by the average of the two values, not by the start.

Slide 8 Two points, one number: percentages of the midpoint

The same two points of the coffee table, 2.00 with 160 cups and 2.50 with 120, give two different elasticities when the percentages are taken from the start: 1.0 going up, 1.67 going down, because a percentage depends on its base. The midpoint method removes the ambiguity: divide each change by the average of the two values. Here 0.50 over 2.25 and 40 over 140, so 22 and 29 per cent, and the elasticity is 1.29 in either direction.

In this course an elasticity between two points always means the midpoint method; the problem sets use it.

One straight line: elastic at the top, inelastic at the bottom



Same slope everywhere; the percentages change:
few cups and a high price at the top, many cups and a low price at the bottom.

Slide 9 One straight line: elastic at the top, inelastic at the bottom

The whole coffee table on its line. Every step of 0.50 in price costs 40 cups, the same slope everywhere, yet the elasticity runs from 0.45 at the bottom to 2.2 at the top. What changes is the base of the percentages: low on the line, 40 cups is a small share of many cups and 0.50 a large share of a low price, so the quantity moves less than the price; high on the line it is the reverse. Exactly in the middle, 160 cups at 2.00, the elasticity is 1.

So a single line does not have an elasticity; it has every elasticity. "Steep means inelastic" is a rule for comparing two lines through one point, not for reading one line.

In pairs, four minutes: compute, classify, and one you cannot

Elastic, inelastic — or no number?

1. Bikinis at the bars: 5.00 → 5.50 euros, 200 → 150 a day
2. A single metro fare +10 %; rush-hour journeys −2 %
3. The T-usual halved in Sep 2022; all journeys in 2023 +17 %

Midpoint method where you have two points; then hands per item.

Slide 10 In pairs, four minutes: compute, classify, and one you cannot

Exercise 1: three price changes and three quantity changes. Before class, decide for each item whether an elasticity can be computed; where it can, use the midpoint method and classify the result. Ask of each item whether its two numbers belong to the same demand curve.

Revealed in class.

Demand is elastic when buyers can walk away

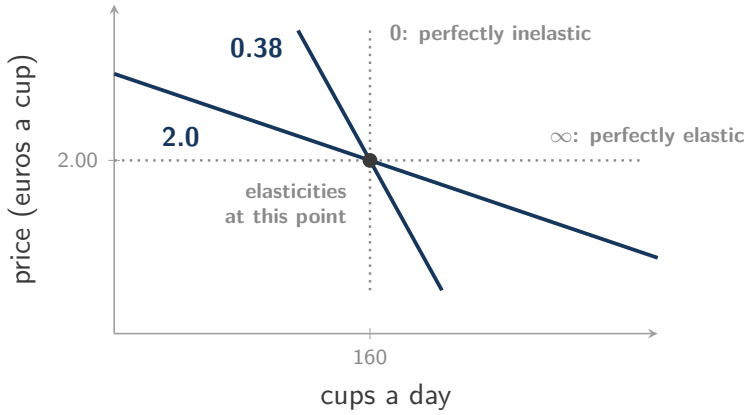
- Close substitutes nearby: Bar Nou, or the bar next door
- A luxury, not a necessity: a dinner out, not bread
- A narrow market: one bar's cup, not "coffee"
- Time to adjust: a year, not a week
- A big slice of the budget: rent, not chewing gum

Slide 11 Demand is elastic when buyers can walk away

The reading's five determinants in one sentence: demand is elastic when buyers can walk away, and walking away is easy when there is somewhere to go. Close substitutes are somewhere to go; a necessity such as bread or the morning metro gives nowhere. The narrower the market, the more substitutes lie outside it: one bar's cup is elastic, coffee less so, hot drinks less still. Time adds options: in a week you keep your commute, in a year you can change it. A big slice of the budget makes you look around; a small one does not.

The common error is to think that expensive goods are inelastic. What matters is the alternatives and the share of the budget, not the price tag.

Through one point: the flatter the line, the more elastic



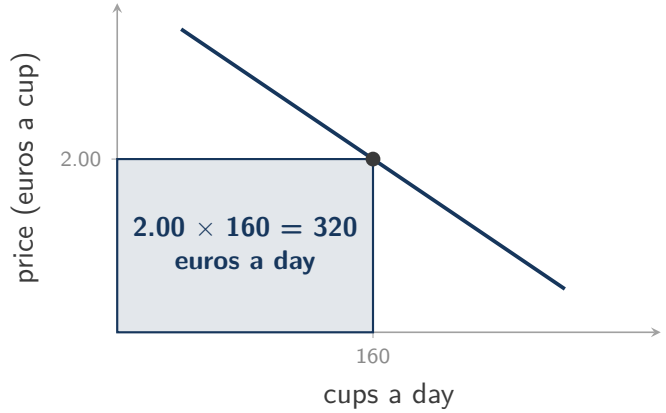
Same axes only. A line that looks flat on a stretched axis means nothing.

Slide 12 Through one point: the flatter the line, the more elastic

Two demand curves through the same point on the same axes. The flat one has elasticity 2.0 there, the steep one 0.38: the same price rise costs more than five times as many cups on the flat line. The extremes have names: vertical, the quantity never moves, perfectly inelastic, elasticity zero; horizontal, perfectly elastic, buyers take any amount at one price and none above it, Class 5's price taker seen from the seller's side.

The warning is the point of the slide: the comparison works only through one point on the same axes. Stretch the price axis and any line looks flat. Read the number, not the look.

What buyers spend is what sellers get: price × quantity, a box



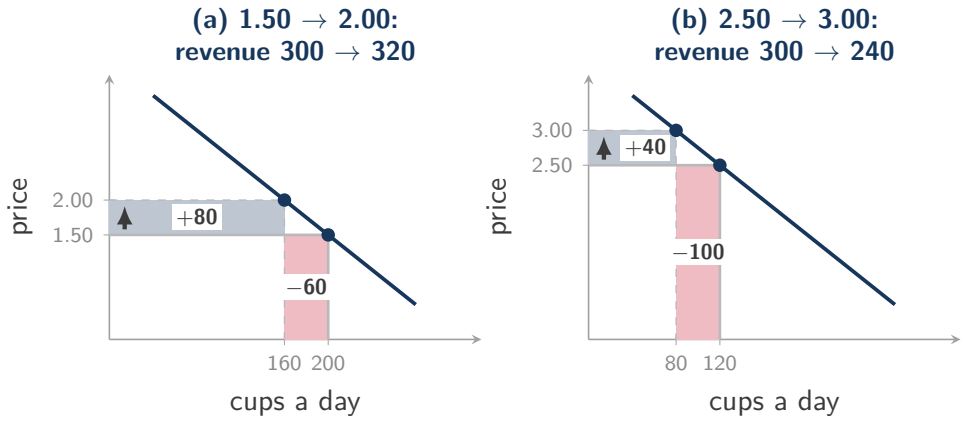
Total expenditure of buyers = **total revenue** of sellers: one box, two names.

Slide 13 What buyers spend is what sellers get: price × quantity, a box

The money in a market is the price times the quantity: 2.00 times 160 cups is 320 euros a day, the box under the point on the demand curve, as tall as the price and as wide as the quantity. Buyers call that box their spending and sellers their revenue; it is one box with two names. Revenue is not profit: the beans and the barista are paid out of it later.

The slide sets up the bar owner's question: raise the price and the box gets taller but narrower. Which wins is the next slide.

A dearer cup: more money at the bottom of the line, less at the top



Inelastic stretch (0.78): the gained strip wins. Elastic stretch (2.2): the lost one does.

Slide 14 A dearer cup: more money at the bottom of the line, less at the top

Two price rises of 0.50 on the same line. In (a), low on the line where the elasticity is 0.78, the bars gain 0.50 on each of the 160 cups they still sell and lose 40 cups that were worth 1.50 each: the gained strip beats the lost one and revenue rises from 300 to 320. In (b), high on the line where the elasticity is 2.2, the gain is 0.50 on only 80 cups while the 40 lost cups were worth 2.50 each: the lost strip wins and revenue falls from 300 to 240.

Same slope, same 0.50, opposite answers, because the cups that walk away were worth different amounts. The box is biggest in the middle, 320 at 2.00, exactly where the elasticity is 1.

Raise the price when demand is inelastic; cut it when elastic

demand	price up	price down
inelastic (< 1)	revenue up	revenue down
unit ($= 1$)	unchanged	unchanged
elastic (> 1)	revenue down	revenue up

Inelastic: revenue follows the price. Elastic: it runs the other way.

Slide 15 Raise the price when demand is inelastic; cut it when elastic

The last two slides as a table. When demand is inelastic, price and revenue move together: a rise brings more money, a cut less. When demand is elastic they move apart: a rise loses money, a cut brings more. At unit elasticity the box does not change.

For a seller who sets a price, a metro authority, a cinema, a bar, this is the first thing to know. For a whole market of price takers, like the olive-oil growers of Class 6, the same rule says what a smaller harvest does to the sector's money, which is the puzzle waiting for the second half.

Which curve moves, and how many members? (optional)

Gym memberships in a city; price elasticity of demand 1.5

1. Memberships get 10 % dearer
2. Workout apps get dearer
3. Gyms' electricity bills double

In pairs, two minutes; then hands per item.

Slide 16 Which curve moves, and how many members? (optional)

Optional. Before class, decide for each event which curve moves and which way, or whether buyers only move along a curve; for the first event, use the elasticity to say how many members are lost.

Revealed in class.

Break

Ten minutes

slide 17 of 30

Slide 17 Break

Ten minutes. After it: the project and Problem set 1, then the olive oil gets its number.

Today: your topic is fixed; Problem set 1 is out

- Project: each group's question is settled today
- Not settled? See me after class
- Problem set 1 (Blocks 1–2): out today
- Due Thursday 22 October

slide 18 of 30

Slide 18 Today: your topic is fixed; Problem set 1 is out

The two practical points of the day. Each project group's question is settled today: the one agreed at the idea meeting, or by e-mail since. A group whose question is still open sees me after class, two minutes. Problem set 1 covers Blocks 1 and 2, everything up to today, and is out today; it is due on Thursday 22 October, in class. The QR code opens the course page.

Olive oil abroad: what the numbers say

Revealed in class

Before class: slide 6's two pairs of numbers, the midpoint method — what elasticity do they give, and what does it say about the money?

Slide 19 Olive oil abroad: what the numbers say

Revealed in class.

Beforehand: take the two pairs of numbers on the hook slide, tonnes and euros in 2022 and 2023, and work out the average price per kilo in each year. Then apply the midpoint method to the two points and ask what the result says about why the money came out the way it did.

In pairs: raise the price, or cut it?

More money, or less?

1. Rush-hour metro, elasticity 0.2: a dearer fare
2. Cinema on a Tuesday afternoon, 2.5: a cheaper ticket
3. Bar Nou alone among twelve bars, 4.0: a dearer cup
4. The neighbourhood's cups: 2.00 → 2.50, 160 → 120 a day

Three minutes; then hands per item.

Slide 20 In pairs: raise the price, or cut it?

Exercise 2: four sellers, each with the elasticity of the demand they face, and a price move. Before class, use the revenue table to say for each whether the money goes up or down; for the neighbourhood's coffee line, compute the revenue before and after and check it against the elasticity of that stretch.

Revealed in class.

Income elasticity: more income, more of most goods, less of some

$$\text{income elasticity of demand} = \frac{\%\text{ change in quantity demanded}}{\%\text{ change in income}}$$

- **Normal good**: positive — a dinner out, a flat white
- **Inferior good**: negative — instant noodles, the cheapest own brand
- Among normal goods: luxuries above 1, necessities below 1

Class 5's 'income' shifter, now with a sign and a size.

Slide 21 Income elasticity: more income, more of most goods, less of some

The same ratio with income in the denominator; this time the sign is kept, because the sign is the classification. Positive: income up, quantity up, a normal good, most things. Negative: income up, quantity down, an inferior good, the things bought because money is short, instant noodles, the cheapest own brand. Among normal goods the size sorts them: below 1 a necessity, like food as a whole, whose share of income falls as income rises; above 1 a luxury, like dinners out or concert tickets.

Class 5's income shifter moved the demand curve right or left; this number says how far.

Cross-price elasticity: substitutes positive, complements negative

$$\text{cross-price elasticity} = \frac{\%\text{ change in quantity of A}}{\%\text{ change in price of B}}$$

- The T-usual half price → fewer T-casual journeys: **substitutes**, positive
- Croissants dearer → fewer cafès amb llet: **complements**, negative

Zero: unrelated goods. Class 5's shift of demand, measured.

Slide 22 Cross-price elasticity: substitutes positive, complements negative

The quantity of one good over the price of another. Substitutes: the other good gets dearer and buyers move to this one, same direction, positive; Class 5's payoff was exactly this, the T-usual cheaper and T-casual journeys down. Complements: croissants dearer, fewer cafès amb llet with them, opposite directions, negative. Unrelated goods: zero.

The sign is the definition; the size says how close the substitutes are: one bar's cup against the bar next door is a large positive number, coffee against tea a small one. The usual slip is to read "both went up" as complements: look at what went up, a price and a quantity of two different goods.

In pairs: which elasticity, what sign, what word?

Name it, compute it, give the word.

1. Income +10 %; instant noodles −5 %
2. Income +10 %; concert tickets +15 %
3. Tea +20 %; cups of coffee +4 %
4. Croissants +10 %; cafès amb llet −3 %
5. A cup +10 %; cups −8 %

Three minutes; then hands per item.

Slide 23 In pairs: which elasticity, what sign, what word?

Exercise 3: five pairs of percentage changes. Before class, decide for each pair which elasticity it describes, which means asking what changed and what responded; then compute it and give the word that goes with its sign and size.

Revealed in class.

Supply elasticity: sellers respond when they can; time is the key

$$\text{price elasticity of supply} = \frac{\%\text{ change in quantity supplied}}{\%\text{ change in price}}$$

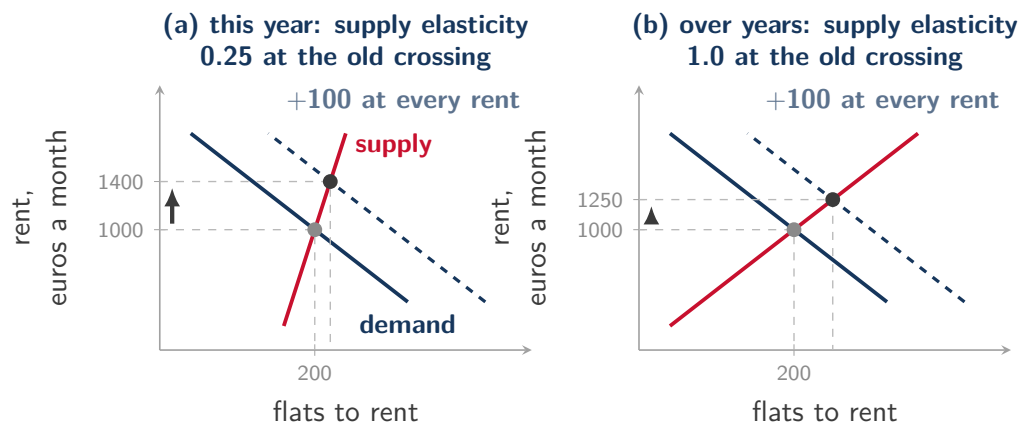
- This year's olives are already on the trees: near 0
- Spare capacity, stock in the warehouse: elastic
- Years to plant, build, hire: more elastic
- Vertical: perfectly inelastic — the seats in a hall tonight

Slide 24 Supply elasticity: sellers respond when they can; time is the key

The same ratio for the other side: how much the quantity supplied responds to the price. The reading's determinants come down to one question: can sellers change the amount, and how soon? This year's olives are already on the trees, so this year's crop cannot answer this year's price: supply near zero elasticity, which is why Class 6's olive-oil supply lines were so steep. A factory at two thirds of capacity or a full warehouse can respond at once: elastic.

Over years anyone can plant, build, hire or enter, so supply gets flatter with time. The extreme is the seats in a hall tonight: vertical, elasticity zero.

Flats in Gràcia: more people want one; this year the rent jumps



Same shift. Steep supply: rent +400, flats +20. Flatter: rent +250, flats +50.

Own numbers; not data.

Slide 25 Flats in Gràcia: more people want one; this year the rent jumps

Own numbers for a street in Gràcia. Demand for flats shifts right by 100 at every rent: students in September, remote workers, tourists. In (a), this year, the stock of flats is what it is and a higher rent brings only a few more out, supply elasticity 0.25 at the old crossing: the rent jumps from 1,000 to 1,400 and just 20 flats appear, so the shock goes almost entirely into the price. In (b), over years, buildings go up and owners rent out, supply elasticity 1: the same shift raises the rent to 1,250 and adds 50 flats.

The elasticity of supply decides how much of a demand shift becomes price and how much becomes quantity. When a rent jumps like this, governments cap it; what a cap does to a market with steep supply is Class 9.

The same shift, with numbers

Revealed in class

Before class: slide 5 once more — in which panel does the price jump more, and in which does the quantity fall more? Say why, given that the shift is the same.

Slide 26 The same shift, with numbers

Slide 5 with numbers. Before class, decide in which panel the price jumps more and in which the quantity falls more, and say why, given that the shift is the same.

Revealed in class.

Two juice stalls: who should make what? (optional)

In one hour: Red stall, 6 smoothies or 3 sandwiches; Blue stall, 2 or 2

1. Red is faster at both. Can trade still help?
2. Who makes the sandwiches?
3. A sandwich price both accept?
4. Blue's hour becomes 2 smoothies or 1 sandwich. Now?

Slide 27 Two juice stalls: who should make what? (optional)

Optional. Before class, work out what a sandwich costs each of them in smoothies, then answer the four questions with those two numbers, not with who is faster.

Revealed in class.

What moves on the frontier? (optional)

A region grows almonds and makes bicycles

1. A better almond variety doubles the harvest
2. Laid-off bicycle workers get their jobs back
3. 5,000 workers move in, able to do either job

In pairs; draw it first.

Slide 28 What moves on the frontier? (optional)

Optional. Before class, draw the region's frontier, almonds on one axis and bicycles on the other, and for each event decide whether the frontier moves, at one end or both, or whether the region moves relative to it.

Revealed in class.

In one line

Revealed in class

Before class: try to say the whole class in two lines — what elasticity measures, and what it decides when supply shifts.

slide 29 of 30

Before Monday 19 October

- Block 3 begins: surplus, efficiency, taxes and subsidies
- Read Mankiw & Taylor, Ch. 6, pp. 141–158 — consumer and producer surplus, market efficiency; Ch. 7, pp. 163–171 — taxes and subsidies
- Problem set 1 (Blocks 1–2): due Thursday 22 October
- Practise in Sextant: the units the course page lists

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slide 30 of 30

Slide 29 In one line

Revealed in class.

Beforehand: try to say the whole class in two lines: what elasticity measures, and what it decides when supply shifts.

Slide 30 Before Monday 19 October

Monday 19 October opens Block 3: what a market is worth to buyers and sellers, surplus and efficiency, and what taxes and subsidies do to it. Reading: Mankiw & Taylor chapter 6, pages 141 to 158, on consumer and producer surplus and market efficiency, and chapter 7, pages 163 to 171, on taxes and subsidies. Problem set 1, on Blocks 1 and 2, is due on Thursday 22 October, in class. Practise in Sextant with the units the course page lists.

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